APPLIED SCIENCE UNIVERSITY
(ASU)
FACULTY OF INFORMATION TECHNOLOGY
(FIT)
MIDTERM EXAM 2023-1

جامعة العلوم التطبيقية الخاصة
كلية تكنولوجيا المعلومات
امتحان منتصف الفصل 2023-1

Subject: *Object Oriented Programming 1- 1301108*

Instructor: Dr.SharefaMurad , Dr.Shatha Awawdeh. I.Baraa Hamada &I.Qaddoumi

Dec. 13, 2023
Time: 2:00
Period: 60 Minutes

Student Name: [redacted]
Student ID: [redacted]

☐ CS ☐ SE ☑ CCC ☐ DSAI ☐ Other

PLEASE NOTE THAT THIS EXAM IS 5 PAGES AND WITH A TOTAL OF 120 POINTS=30 MARKS.

QUESTION 1: MCQ: Select the correct answer **[CO: K1]** **[24 POINTS]**

1. **[4 points]** Which of the following statements is true about a default constructor?

a)	It must be explicitly defined in each class.
b)	It has parameters provided by the programmer.
c)	It is automatically provided by the compiler if no constructor is defined.
d)	It cannot be used to initialize variables

2. **[4 points]** The value of : **System.out.println((int) Math.ceil(3.6));** is:

a)	3.0	b)	4
c)	3	d)	4.0

3. **[4 points]** Which of the following can be placed in the blank line in the following code?

```
public class Test {
 private int id;
 public void m1() {
  _____.id = 45;
}}
```

a)	this
b)	Test
c)	super
d)	This

4. **[4 points]** Which of the following expressions yields an integer between [15 and 45], inclusive?

a)	15 + (int)(Math.random() * 31)	b)	15 + (int)(Math.random() * 45)
c)	(int)(Math.random() * 45)	d)	15 + (int)(Math.random() * 30)

15+31 = 46 [15 - 46]

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TION 1: [6 POINTS]

the correct answer.

A. **[1 POINT]** The procedural/structured paradigm focuses on functions, whereas the object-oriented paradigm combines data and methods into objects.

a. True

b. False

B. **[1 POINT]** nextChar() method in the Scanner class is used to read a character in Java.

a. True

b. False

C. **[1 POINT]** In Java, a constructor must always have the same name as the class and can return a value.

a. True

b. False

D. **[1 POINT]** The _____ access modifier makes members accessible from everywhere within the java project.

a. static

b. private

c. public

d. Default

E. **[1 POINT]** The process in Java that allows defining two or more methods within the same class that share the same name, as long as their parameters are different, is:

a. Inheritance.

b. Encapsulation.

c. Overloading.

d. Polymorphism.

F. **[1 POINT]** Which keyword is used to define a **constant** in Java ?

a. final

b. static

c. constant

d. private

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TION 2: Evaluate the following Java codes and show the exact output. **[4.5 POINTS]**

A. **[2 points]**

```
public class NewClass {
    static int Count = 0;
    private int x;
    public NewClass() {
        this(1);
        ++Count;
    }
    public NewClass(int i) {
        x = i;
        ++Count;
    }
    public String toString() {
        return x + " " + Count;
    }
}
```

```
public class Main {
    public static void main(String[] args) {
        System.out.println(NewClass.Count);
        NewClass q1 = new NewClass();
        NewClass q3;
        System.out.println(NewClass.Count);
        q3 = q1;
        System.out.println(q3);
    }
}
```

Count	x	q1	q3
0	0	0	0
1		1	1

B. **[1 point]**

```
public static void main(String[] args) {
    double x=1+Math.random()*10;
        if(x<11)
            System.out.println("welcome to java");
        else
            System.out.println("I love java");
}
```

welcome to j-va

C. **[1.5 points]**

```
public static void main(String[] args) {
        String s1="Hello ";
        String s2="oops";
        String s3=s1.charAt(4)+s2.substring(1,3)+" 1";
        System.out.println(s3);
        String s4=s1.concat(s3);
        System.out.println(s4);
        System.out.println(s4.indexOf('h'));
    }
```

oop1

Hello oop1

-1

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QUESTION 3. Analyze each line of the following code and write in the table below whether the marked lines have an error or not. If the line has an error, then explain why. **[5.5 POINTS]**

```
public class Vehicle {
   private int wheels;
   private int doors;
   private int seats;
   public Vehicle() {
      this.wheels = 4;  ----------------------------------------------------------> //line 1

      doors = 4;
   }

   public static Vehicle(int wheels, int doors, int seats) { ----------------> //line 2
      this.wheels = wheels;
      this.doors = doors;
      this.seats = seats;
   }

   public static boolean canCarry(int numberOfPeople) {
      return (seats >= numberOfPeople); ----------------------------------> //line 3
   }
}
```

```
public class TestVehicle {
   public static void main(String[] args) {
      Vehicle car1 = new Vehicle(); ----------------------------------------> //Line 4

      Vehicle car2 = new Vehicle(4); ---------------------------------------> //Line 5

      Vehicle car3 = new Vehicle(4, 2, 5);

      car1.doors = 2; ------------------------------------------------------> //Line 6

      System.out.println(car1); --------------------------------------------> //Line 7
   }
}
```

Line	Correct/Incorrect	Explanation why if incorrect
Line 1	✓ ✓	—
Line 2	✗ ✓	cannot have a static constructor ✓
Line 3	✓ ✗	—
Line 4	✓ ✓	—
Line 5	✗ ✓	~~not enough parameters~~ ✓ Not enough parameters | Vehicle car2 = new Vehicle (4, 4, 4);
Line 6	✗ ✓	✓ doors is private and can't be accessed by setters only
Line 7	✓ ✓	—

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TION 4: [14 POINTS]

ly object-oriented programming principles to develop a solution for the given problem:

A. [5.25 POINTS] Write a java class named **QuestionClass**. This class contains the following:

- A private instance variable: **questionText** of type String, which represents the text of the question.
- A parameterized constructor with one parameter to initialize **questionText**.
- A public method **getQuestionText** to return the instance variable of the class.
- A public method **setQuestionText** that assigns text to the question. This method includes validation to ensure that the text is not null. If the text is found to be null, the method will store **"No question provided"**. if (textlength() <=0)

B. [5.25 POINTS] Write a java class named **Assignment** that contains the following:

- Two instance variables:
  - A private **assignmentTitle:** of type String.
  - A private **question**: This instance variable should be a composition of the **QuestionClass**. new
- A parameterized constructor with two parameters to initialize **assignmentTitle** and **questionText** in **question**.
- A **toString()** method that returns the details of the **assignmentTitle** and **questionText** from the **question**.

C. [3.5 POINTS] Create a Java class called **Test** and write a main method that performs the following actions:

- Ask the user to enter details of the Assignment, including the **assignmentTitle** and **questionText**.
- Declare an object of **Assignment** type with the values entered by the user: **assignmentTitle** and **questionText**.
- Print all the information of the Assignment object.

~~public class Test {~~

~~public static void main (String[] args) {~~

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20

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5. **[4 points]** Assume the variable s is " abc ", the method ____ returns a new string "abc".

a)	s.trim(s)	b)	String.trim(s)
c)	trim(s)	d)	s.trim()

6. **[4 points]** What is the main purpose of a static data field?

a)	To store unique values to each instance of a class	b)	To provide default values for instance variables
c)	To share a value among all instances of a class.	d)	To prevent other classes from accessing the field Private

QUESTION 2: Trace the following java code fragments and write down their outputs?

[CO:S1,SO :1.2] **[14 POINTS]**

A. [8 points]

```
public class Demo {

public static void main(String[] args) {
 String s1="JORDAN";
 String s2="VA LID";
 String s3=s1.charAt(0)+"A"+s2.substring(0,2);
 s1=s1. toLowerCase();
 System.out.println(s3);
 System.out.println(s1);
 System.out.println(s1.indexOf("A"));
 System.out.println(s3.lastIndextOf("A"));
  }
}
```

JAVA
JAVA
jordan
4
3

6

B. [6 points]

```
public class Main {

public static void main(String[] args) {
      String w1 = "Java";
      String w2 = "Java";
      String w3 = "JAVA";
      String w4 = new String("Java");
      System.out.println(w1==w2);
      System.out.println(w1==w4);
      System.out.println(w1.equals(w3));

  }
}
```

True
True
False

4

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QUESTION 3: [co: s2, so:1.1] [co :s2,so :2.2] 81 [82 POINTS]

Having the following code of a BankAccount class, read it carefully then do the following:

```
public class BankAccount {
    private int accountType;
    private double balance;
    private String customerName;

public BankAccount(int accountType, double balance, String
customerName) {

        this.accountType = accountType;
        this.balance = balance;
        this.customerName = customerName;
    }

    public double getBalance() {
        return balance;
    }
    public void setBalance(double balance) {
        this.balance = balance;
    }
public String toString() {
return "accountType=" + accountType + ", balance=" +
balance + ", customerName=" + customerName;
    }
```

A. Write a java class named **Employee** that contains the following:

- Two attributes (instance variables):
  - A private **name: String**, cannot be empty (number of characters must be greater than zero) otherwise set the name to "No Name".
  - A private **account: BankAccount**, should be set as an **aggregation** of **BankAccount** class.
- A parameterized constructor with two parameters to initialize **name** and **account** in the **Employee** class.
- Two accessor methods (getters) for name and account attributes.
- Two mutator methods (setters) for name and account attributes.
- A **toString()** method that returns the details of the Employee **name** and **account** in the Employee.

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```
Public class Test() {
    Public static void main (String [] args) {
~~Employee obj1 = new Employee ("Mohannad",~~
BankAccount x = new BankAccount (2, 1000, "Ali);
Employee obj1 = new Employee ("Omar", x);
    obj1.setBalance (1300);
System.out.Println (obj1);
  }
}
```

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B. Create a Java class called **Test** and write a main method that performs the following actions:

- Create an Employee object called **obj1** using a parameterized constructor.
- Change the balance of **obj1** to 1300.
- Print the information of the Employee **obj1**.

```
Public class Employee () {
    Private String name;
    Private BankAccount ref= BankAccount;

Public Employee (String name, BankAccount account)
  {
    if(name.length() > 0)
      {
        this.name = name;
      }
    else
      {this.name = "No Name";
      }
    this.account = account;
  }

Public String getName ()
  {
    return name;
  }

Public void setName (String name)
  {if (name.length() > 0)
      this.name = name;
    else
      this.name = "No Name";
  }

Public BankAccount getAccount()
  {
    return account;
  }

Public void setAccount (BankAccount a)
  {
    this.account = a;

Public String toString () {
    return "Name : " + this.name + "\n Account : "
    + this.account;
  }
}
```


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